

ARRAY

Definition:

- Array is the collection of multiple data stored in a single variable.
- Array store elements in continuous memory location.
- It stores only same data type elements.
- In Array elements accessed with index which start from 0 to Array length – 1.
- Array is an Object.

Declaration:

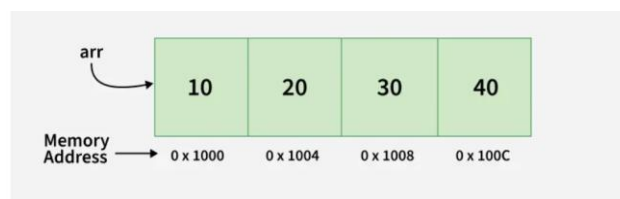
Syntax: `datatype Variable_name[] = new datatype[array_size];`

Example:

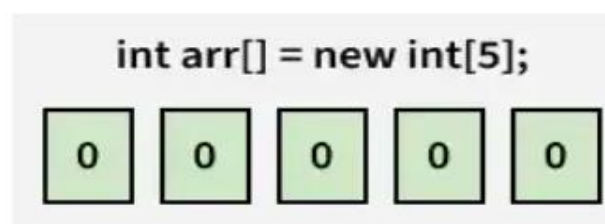
Declaring array runtime input - `int arr[] = new int[5];`

Declaring array compile time input – `int arr[] = {4,5,6,7,8};`

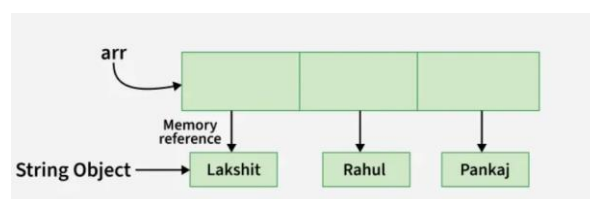
Representation in memory:



Primitive Data Type



When declaring initially for all data Zero is Stored



Non-Primitive Data Type

ADVANTAGES:

- Fast access using index.
- Easy to traverse.
- Less memory overhead.
- Simple implementation.

DISADVANTAGES:

- Similar data type only stored.
- Fixed size.
- Lack of inbuilt methods.